

established with a particular pincode.

2.5.5.5. CASE Authenticated Tag

This subrange of Node ID is used to assign an access control subject to a group of peer nodes that share a single CASE session as specified in [Section 6.6.2.1.2, “Subjects identified by CASE Authenticated Tag”](#).

2.5.5.6. Unspecified Node ID

The Unspecified Node ID (0x0000_0000_0000_0000) is a reserved value that never appears in messages or protocol usage. It exists to mark or detect the presence of uninitialized, missing, or invalid Node IDs.

2.5.6. IPv6 Addressing

This protocol uses IPv6 addressing for its operational communication. Node IDs and Fabric IDs are resolved to various types of IPv6 addresses [[RFC 4291](#)].

2.5.6.1. IPv6 Unicast Address

An IPv6 Unicast Address is one that uniquely identifies and addresses a single Node on an IPv6 network. A primary design goal for this standard is to allow Nodes to leverage native IPv6 technologies. As such, an operational IPv6 Unicast address that provides connectivity and routability between Nodes SHALL be used. This includes a global unicast address (GUA), a link-local address (LLA), or a unique local address (ULA).

2.5.6.2. IPv6 Multicast Address

An IPv6 Multicast Address is formed using Unicast-Prefix-based IPv6 Multicast Addresses [[RFC 3306](#)]:

- The first 12 bits are defined by [[RFC 3306](#)] and are 0xFF3.
- The next 4 bits are "scop" (scope) and set based on [[RFC 7346](#)] Section 2 to:
 - *Site-Local* (0x5) - spans all networks in the Fabric, including Thread, Ethernet, and Wi-Fi.
- The next 8 bits are reserved (0x00).
- The next 8 bits are "plen", and set to 0x40 to indicate a 64-bit long network prefix field.

The network prefix portion of the Multicast Address is the 64-bit bitstring formed by concatenating:

- 0xFD to designate a locally assigned ULA prefix per [[RFC 4193](#)] Section 3.1
- The upper 56-bits of the Fabric ID for the network in big-endian order

The 32-bit group identifier portion of the Multicast Address is the 32-bits formed by:

- The lower 8-bits of the Fabric ID
- 0x00
- The next 16-bits are the Group Identifier for the group, as specified in [Group Identifier](#) in big-